Balaji Leninrajan

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Education

University of Waterloo 2023 – Present
Bachelor of Software Engineering *Waterloo, ON, Canada*
Coursework: Operating Systems, Compilers, Data Structures, Algorithms, Databases, Computer Architecture

Experience

Shopify | Production Engineering Intern | Edge Infrastructure *September 2025 – December 2025*
- Deployed Terraform-based Cloudflare WAF rules to safeguard service migrations, achieving 100% monitoring coverage (up from 96.4%) and ensuring zero-downtime transitions across critical services.
- Accelerated incident resolution by utilizing Prometheus for metrics analysis and documenting recovery procedures in new operational playbooks.
- Extended Shopify's Cloudflare Ruby gem to query custom hostnames and metadata.
- Supported migration from Nginx to Envoy, including porting Lua scripts to Go, to improve edge routing.

Shopify | Software Engineering Intern | Communications Platform Team *January 2025 – April 2025*
- Contributed to a high-scale communications platform using React and Rails, orchestrating 4M+ monthly emails and push notifications across multiple Shopify services and teams.
- Implemented an approval workflow system for emails that empowers users in marketing automation, resulting in an 80% reduction in faulty campaigns due to human error.
- Designed a polymorphic Rails concern for approval workflows, abstracting shared logic across multiple models, reducing duplicate code and improving maintainability.
- Optimized error monitoring by implementing intelligent Bugsnag filtering and categorization, reducing non-actionable alerts by 70% and improving response time.

Waterloo Aerial Robotics Group | Autonomy Lead *January 2024 – Present*
- Led a team of 10+ software developers that contributed to back-to-back victories at AEAC 2024 and 2025.
- Increased search area coverage efficiency by developing a multi-drone pathing system in Python with peer-to-peer networking, enabling parallel operations and collaborative area partitioning.
- Improved code reliability and team productivity by establishing robust quality control processes including pytest-based CI/CD pipelines and systematic code reviews.

YTY Group | ERP/Web Development Intern *May 2024 – August 2024*
- Engineered an employee attendance prediction system using TensorFlow and Pandas for data preprocessing, improving HR planning capabilities and reducing administrative workload.
- Created a Flutter-based expense claims mobile app with offline functionality and persistent login, reducing submission time and improving accessibility.

Projects

Maram | *Rust, Git, Zellij*
- Built a Rust CLI that automates git worktree management for parallel feature experimentation.
- Integrates Zellij to instantly provision isolated terminal workspaces for each branch variant.
- Implements logic to diff and cherry-pick changes across worktrees without manual git operations.

Linguini | *TypeScript, React, Next.js, tRPC, Tailwind, PostgreSQL*
- Built a multiplayer word-guessing game with group leaderboards and real-time progress tracking.
- Developed the main game loop on the frontend, ensuring smooth state management and responsive gameplay.
- Implemented tRPC routers for authentication, user management, and group management.

TapCode | *Flutter, Google Gemini*
- Engineered an accessibility-focused application that converts speech to haptic morse code patterns using Flutter's vibration API.
- Integrated Google Gemini for text optimization, reducing transmission time while maintaining message integrity.
- Implemented bi-directional translation between speech and morse code, enabling two-way communication for users with visual or auditory impairments.

Technical Skills

Languages: Python, C, C++, Go, Ruby, Rust, Typescript, Javascript, SQL, Assembly, Scala, Dart, Bash
Technologies: Rails, React, React Native, Next, Flask, GraphQL, gRPC, Tailwind, Nginx, Envoy, Linux Terraform, Git, AWS, RaspberryPi, PyTorch, TensorFlow, Pandas, Kubernetes